Open Set Few-Shot Remote Sensing Scene Classification Based on a Multiorder Graph Convolutional Network and Domain Adaptation

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***Abstract*— Few-shot scene classification aims to recognize task data, which consists of unlabeled data and a few annotated samples, given some labeled auxiliary data. The task and the auxiliary data often come from the same source with nonoverlapping categories; such tasks are defined as closed set few-shot scene classification. However, most remote sensing applications involve open set few-shot scene classification, where the categories of the task and auxiliary data are often partly overlapped and sources of data are usually different (i.e., different domains). We propose an open set few-shot scene classification method called the multiorder graph convolutional network (MGCN) specifically for such tasks. MGCN solves the open set few-shot scene classification task in two ways: reducing the interdomain discrepancy by a feature dispersion degree weighting domain adaptation method and reducing the dispersion degree of node features by a proposed MGCN. Based on a shallow backbone and an improved graph convolutional network, MGCN improves the classification performance of the open set few-shot classification task by utilizing information from common categories and reducing domain discrepancies. The experimental results, based on three public remote sensing image datasets and a Tibetan Plateau remote sensing scene dataset that we collected, demonstrate the effectiveness of the proposed method on the open set few-shot scene classification task.**

***Index Terms*— Domain adaptation (DA), graph convolutional networks (GCNs), open set few-shot classification, remote sensing scene.**

I. Introduction

SCENE classification aims to label remote sensing images with different categories according to the contents of the images. It is broadly applied in reality, such as urban planning, environmental protection, and land cover [1], [2]. Deep learning methods have been successfully applied in the field of remote sensing scene classification [3], [4]. However, existing deep learning methods generally require large numbers of labeled samples for training, and obtaining enough annotated remote sensing images is labor-intensive and time-consuming, making it unrealistic in many cases. In this regard, scholars have studied few-shot classification to avoid the problem of insufficient labeled samples in remote sensing scene classification [5], [6].

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Few-shot classification aims to learn a classifier by a labeled auxiliary dataset D_A and a few labeled samples in a task dataset D_T and then recognize those unlabeled samples in D_T. D_A and D_T usually come from different sources. In the existing works [5], [6], [7], it is generally assumed that, as shown in Fig. 1(a), the label sets in D_T and D_A are orthogonal, that is, $Y_T \cap Y_A = \phi$, where Y_T and Y_A denote the label sets of D_T and D_A. The set of labeled samples in D_T is called the support set, and the set of unlabeled samples is called the query set.

However, in practical applications, the label sets Y_T and Y_A may be nonorthogonal, i.e., $Y_T \cap Y_A \neq \phi$ and $Y_T \cap Y_A \neq Y_T$, just as shown in Fig. 1(b). To distinguish between these two few-shot classification tasks, we refer to the few-shot classification task in the nonorthogonal case as open set few-shot classification and that in the orthogonal case as closed set few-shot classification. In addition, in the nonorthogonal case, the overlapping categories are called common categories, and the rest are called private categories.

For closed set few-shot classification, many approaches have been proposed. For example, Snell et al. [8] proposed a metric-based method called prototypical network (ProtoNet). ProtoNet maps samples in each category into a feature space, extracts the "mean" of the samples to represent the prototype of the category, and then predicts the category of unlabeled samples based on the distance between unlabeled samples and each prototype in the feature space. Ma et al. [9] proposed a transductive relation-propagation method, which can predict the category of unlabeled samples through the semisupervised classification of label propagation. Ravi and Larochelle [10] used auxiliary data to train a generalized classification model and then fine-tuned it with a small number of labeled samples from D_T to obtain a classification model on the task data.

We can directly adopt the existing closed set methods for the open set few-shot classification. However, different source images differ in many aspects, such as sensors, camera perspectives, geographic locations, seasons, and weather conditions, the feature distributions of D_A and D_T may be different apparently. Due to the discrepancy in distributions of image features between the two domains, models trained based on D_A often cannot achieve high classification accuracy on D_T even for the common categories. In the open set case, because of the existence of common categories, we can obtain the distribution discrepancy between D_A and D_T.

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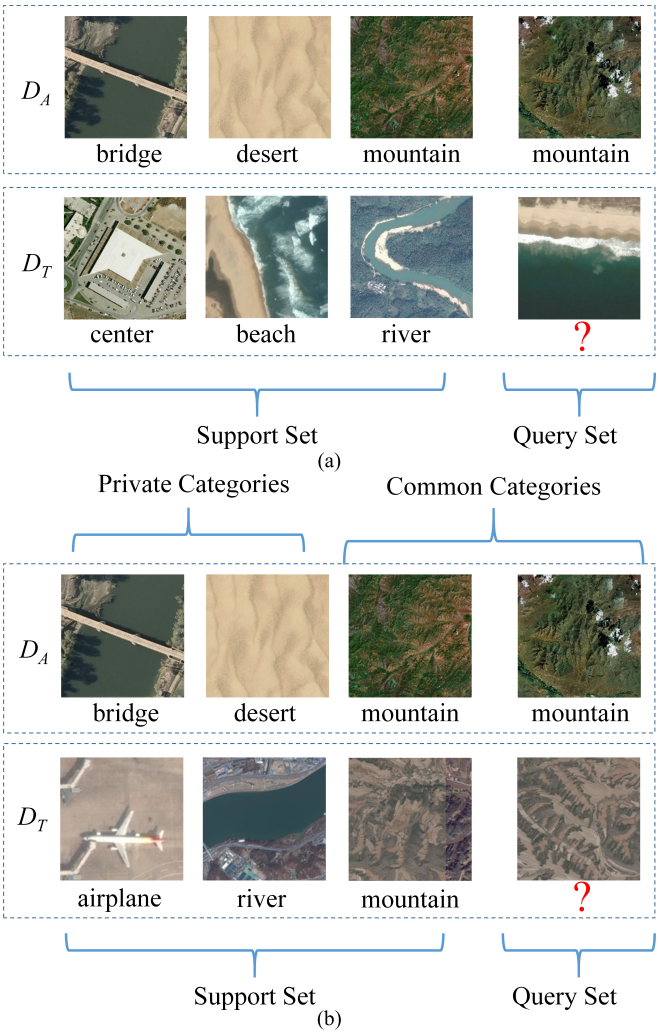


Fig. 1. Different classification tasks. (a) Closed set few-shot classification. (b) Open set few-shot classification.

We presume that the classification accuracy may improve when reducing domain discrepancy even by existing few-shot classification methods. Furthermore, the existing few-shot classification methods rarely consider the relationships among samples. If we adopt a graph to construct the relationships among samples and update the image features by graph learning, we can obtain feature representations that are conducive to classification, which can further improve classification accuracy.

To this end, we propose a framework called the MGCN for open set few-shot remote sensing scene classification based on ProtoNet. First, we introduce a domain adaptation (DA) strategy to ProtoNet. DA quantifies the discrepancy between two datasets as a domain loss function and reduces the discrepancy by minimizing this loss function during training. To improve the effectiveness of DA, we propose a dispersion degree weighting DA (WDA). The proposed WDA assigns larger weights to the domain loss of the categories with high feature dispersion and smaller weights to the domain loss of the categories with low feature dispersion; thus, WDA can focus on the hard-to-distinguish categories to reduce domain discrepancies more effectively. Second, we construct a graph convolutional network (GCN) based on the similarity among image features, propose a multiorder graph convolution to reduce the dispersion of sample features, and, thus, further improve the classification accuracy. We evaluate the proposed approach based on three public benchmark datasets and one collected Tibetan Plateau dataset. The experimental results show that the proposed MGCN outperforms the compared methods dramatically in cases when only one or five target samples are labeled in each category.

In summary, the contributions of our work are threefold.

1) We propose a novel image classification method MGCN to solve the few-shot classification problem when label sets of the auxiliary data and the task data partially intersect. To the best of our knowledge, this is the first attempt to integrate the idea of DA to address the open set few-shot classification task specifically.
2) We find that DA based on common categories can reduce the category feature dispersion degree. Based on this, we propose a feature dispersion degree weighting domain adaptation method called WDA. WDA can enhance the effectiveness of DA and improve classification accuracy by assigning large weights to the categories with high feature dispersion degrees and small weights to those with low feature dispersion degrees in the training phase.
3) We further utilize a GCN to model the feature similarity among images and propose the multiorder graph convolution. Through theoretical analysis, multiorder graph convolution can enhance the feature smoothing ability adaptively based on learned parameters compared with existing graph convolution and, thus, can further improve the classification accuracy.

The remainder of this article is organized as follows. In Section II, we introduce works related to few-shot image classification, DA, and GCNs. Section III describes the proposed method. We present the experimental results in Section IV. Finally, discussions and conclusions are given in Section V.

II. Related Works

In this section, we review the works related to remote sensing scene classification, few-shot learning, DA, and GCNs.

A. Remote Sensing Scene Classification

In recent years, deep learning has been successfully applied in various fields and has increasingly prevailed in the area of remote sensing image scene classification due to its excellent feature extraction capabilities [11]. These methods are mainly based on a convolutional neural network (CNN) and can be roughly divided into three types: traditional CNN-, fine-tuning-, and DA-based methods. The traditional CNN method mainly solves the scene classification problem when there are sufficient training samples. By using different strategies for exploiting CNN, a variety of CNN-based scene classification methods [11], [12] have emerged. When the amount of training samples is inadequate, fine-tuning an already trained CNN on target datasets is a good choice [13], [14]. In the fine-tuning-based method, a pretrained network is adopted as a

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feature extractor, and parts of the network are fine-tuned with a small number of labeled samples to obtain the classification network. The fine-tuning-based method usually ignores the distribution discrepancy among different datasets. To solve the problem of distribution discrepancy across datasets, DA-based methods are proposed [15], [16]. The main idea of DA-based methods is to quantify the interdomain discrepancy as domain loss and then align the feature distributions by minimizing the domain loss. However, most DA-based methods address the problem when the categories of the training dataset and the test dataset are the same, and little work [17] has been done on the open set task, which is more practical and challenging.

B. Few-Shot Learning

In most cases, scholars set few-shot learning as an N-way K-shot problem [18], where N refers to the number of categories in D_T and K refers to the number of labeled samples in each category in D_T. The purpose of few-shot classification is to recognize unlabeled samples based on few labeled samples and D_A. Scholars have proposed many methods to solve the closed set few-shot classification problem. These methods can be broadly classified as metalearning methods, and metric- and graph-based methods.

The main idea of metalearning methods [19] is that, by training a model based on a large number of different sampled datasets, the model obtains good initial parameters, and for a new learning task, only a small amount of fine-tuning is needed to effectively perform the new task. The well-known methods in metalearning are model-agnostic metalearning (MAML) [20] and Reptile [21]. MAML developed a learning-based model parameter initialization scheme that can be generalized to similar datasets. Based on MAML, Reptile improved the gradient calculation method, which can significantly reduce the complexity of model learning. Metalearning methods generally need a lot of auxiliary data for training; when auxiliary data is not enough, we may not obtain good initial parameters. In addition, there may appear overfitting in the process of fine-tuning [22].

The goal of metric learning is to learn a distance metric function that measures how similar or related two objects are through the auxiliary dataset D_A, and then, the categories of unlabeled samples in D_T can be predicted based on the metric distance between unlabeled and labeled samples. In terms of metric learning, Koch et al. [23] pioneered the integration of deep learning and few-shot learning by using a Siamese CNN to learn similarity measures between samples over sample pairs. Vinyals et al. [24] proposed a matching network that introduces an attention mechanism to recognize unlabeled samples by a K-nearest neighbor (KNN) [25] method based on cosine distance. Li et al. [26] added a fitness loss term to the metric loss function in ProtoNet and improved the classification accuracy by balancing the loss function. The metric-based methods do not cause the overfitting problem compared with metalearning methods. However, this kind of method usually ignores the relationships among samples that may be beneficial to performance enhancing.

The graph-based methods use the graph to construct relationships among samples and predict unlabeled samples through the propagation of labels or features on the graph. Garcia and Bruna [27] first adopted a graph in the few-shot classification task. Specifically, a multilayer perceptron is used to construct a graph among samples, and the features of samples are updated by the propagation of neighbors' features; then, the prediction probability for each unlabeled sample is obtained by the updated feature. Liu et al. [28] used KNN to construct the relationships among image features, and then, label propagation was used to estimate the label of the unlabeled samples. Using a graph to propagate labels or features is actually a smoothing operation on graph signals [29], and the degree of smoothing has an impact on classification accuracy [30], but the above methods ignore this point.

The above methods solve the few-shot classification problem when the label sets of auxiliary data and task data are completely different. In practice, we often encounter open set few-shot classification problems. Although the existing few-shot methods are also applicable in the open set task, direct use of existing methods will ignore the interdomain discrepancy that exists between D_T and D_A, which is difficult to obtain an ideal classification accuracy. To tackle this problem, we propose an open set few-shot classification method based on DA.

C. Domain Adaptation

DA [31], [32] is a research field associated with transfer learning in machine learning. This scenario arises when we aim at learning a well-performing feature extraction model from source data, and the model also well performs for target data. Different from few-shot learning, the label sets of the two domains usually are the same. Generally, DA methods can be divided into two types: discrepancy- and adversarial-based methods. The discrepancy-based method generally aligns the feature distributions of two datasets by minimizing a metric function that quantifies the domain discrepancy between the two datasets, which is equivalent to learning a common feature space of the source and target domains. Thus, the network trained based on a training dataset can also solve the same task for the target dataset. The discrepancy-based methods have achieved encouraging performance for the task of image classification [33], [34], [35], [36], [37].

Recently, great success has been achieved by adversarial-based methods [38], which align different domains via an adversarial process. The domain-adversarial neural network (DANN) was first introduced by Ganin and Victor [39]. The idea is to add a domain-adversarial loss to the network instead of the predefined metric function as the discrepancy-based method does. This enables the network to learn more discriminative information. Following the idea of DANN, many works adopted domain-adversarial training [16], [40], [41], [42].

Existing DA methods rarely consider the distribution of different category features. In order to improve the effectiveness of DA and focus DA on categories that are easy to be misclassified, we propose a DA method based on the category features dispersion weighting degree and prove that it can improve classification accuracy.

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D. Graph Convolutional Networks

The concept of convolution for graph data was first proposed by Bruna et al. [43]. They developed the operation of "graph convolution" based on spectral property, which convolves on the neighborhood of every graph node and produces a node-level output. After that, many extensions of graph convolution have been investigated and achieved advanced results. For instance, Defferrard et al. [44] proposed a formulation of CNN in the context of spectral graph theory. Based on their work, Kipf and Welling [45] proposed a fast approximation localized convolution, which makes GCN able to encode both graph structure and node features. In their work, GCN was simplified by the first-order approximation of graph spectral convolution, which leads to efficient smoothing operations. Li et al. [46] and Wu et al. [47] proposed using the second-order graph convolution instead of the first-order graph convolution in GCN, and demonstrated analytically and experimentally that the proposed graph convolution has a more efficient smoothing ability for node features.

The results in [46] and [47] show that proper smoothing can improve classification accuracy and oversmoothing in graph convolution will deteriorate classification results, but existing works rarely consider how to adjust the smoothing ability of GCN. We propose a multiorder graph convolution that can adaptively enhance the smoothing ability and obtain higher accuracy in the few-shot classification compared with existing methods.

III. Methodology

Based on ProtoNet [8], the proposed MGCN is a brand new method designed for open set few-shot classification tasks by new WDA and multiorder GCNs (MGCNs). In this section, we first briefly introduce ProtoNet and the relationship between feature dispersion degree and classification accuracy in Section III-A. Next, we describe the proposed MGCN and illustrate the motivation of WDA and multiorder graph convolution in Section III-B. Third, we describe the proposed WDA and the multiorder graph convolution in Sections III-C and III-D in detail, respectively. Finally, we introduce the metric module and the training process of the MGCN in Sections III-E and III-F, respectively.

A. Feature Dispersion Degree in ProtoNet

ProtoNet is a few-shot classification method that learns a metric space in which classification can be performed by computing the Euclidean distances to the prototype representations of each category. The illustration of ProtoNet is shown in Fig. 2.

In the few-shot classification, there is a support set of L labeled samples $S = \{(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_L, y_L)\}$, where $\boldsymbol{x}_i$ $(i = 1, 2, \ldots, L)$ is the sampled image, $y_i \in \{1, \ldots, N\}$ is the corresponding label, and N is the number of categories in D_T. ProtoNet computes a prototype $\boldsymbol{c}_k$ for each category through an embedding function f with learnable parameters

$$\boldsymbol{c}_k = \frac{1}{|S(k)|} \sum_{(\boldsymbol{x}_i, y_i) \in S(k)} f(\boldsymbol{x}_i) \tag{1}$$

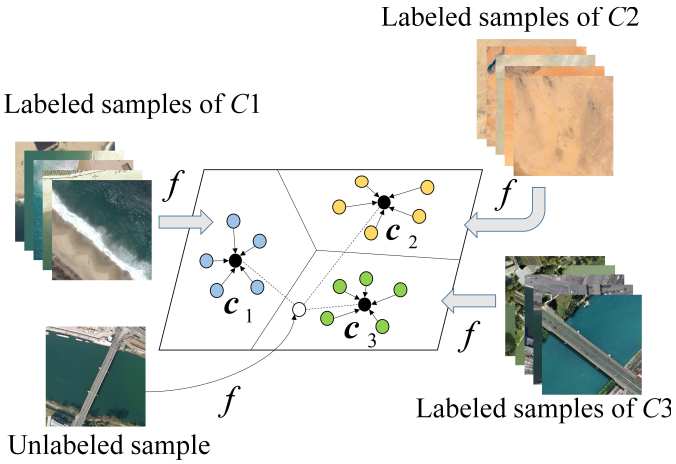


Fig. 2. Illustration of ProtoNet applied for five-shot classification, in which labeled samples are from categories $C1$, $C2$, and $C3$, and one unlabeled sample is embedded in a shared space through an embedding function f. Here, $\boldsymbol{c}_1$, $\boldsymbol{c}_2$, and $\boldsymbol{c}_3$ are the prototypes for categories $C1$, $C2$, and $C3$.

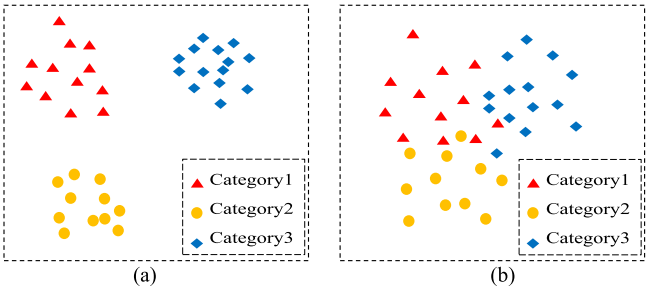


Fig. 3. Illustration of feature distribution with different dispersions. (a) Intracategory variance of each category is small, and the intercategory distance is large. (b) Intracategory variance of each category is large, and the intercategory distance is small.

where $S(k)$ denotes the set consisting of images with labels of the kth category and $|S(k)|$ indicates the number of elements in category k. For an unlabeled sample $\boldsymbol{x}$, ProtoNet produces a prediction probability over categories based on the softmax function

$$p(y = k|\boldsymbol{x}) = \frac{\exp(-\|f(\boldsymbol{x}) - \boldsymbol{c}_k\|_2)}{\sum_i \exp(-\|f(\boldsymbol{x}) - \boldsymbol{c}_i\|_2)}. \tag{2}$$

For ProtoNet, if the intracategory variance of each category is small and the intercategory distance is large, just as shown in Fig. 3(a), the classification accuracy will be comparatively high. Conversely, if the intracategory variance of each category is large and the intercategory distance is small, just as shown in Fig. 3(b), the classification accuracy will be comparatively low. In other words, the degree of feature dispersion of categories has an important influence on the classification accuracy of ProtoNet.

The feature dispersion degree is determined by two aspects: one is the dispersion degree within each category (which can be measured by variance), and the other is the distance between categories' centers. Thus, we use the ratio between the intracategory variance S_c and the square of the nearest neighbor category distance $\hat{d}_c^2$ in the feature space to quantitatively measure the cth category feature dispersion degree, and

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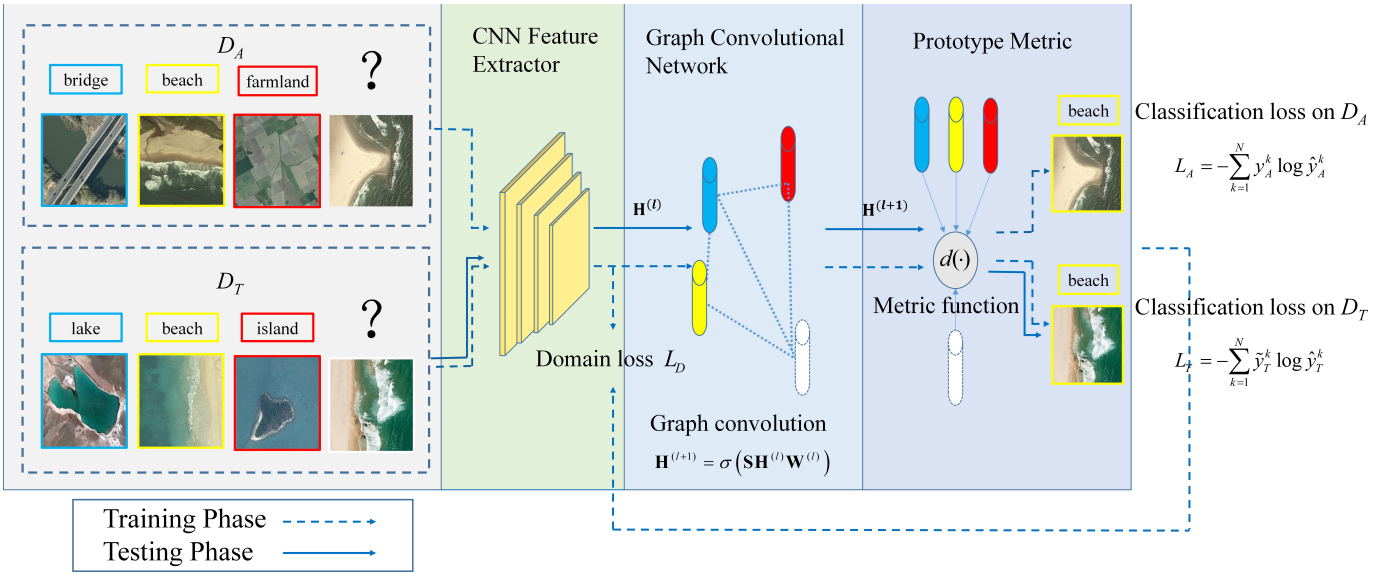


Fig. 4. Framework of the proposed MGCN for few-shot remote sensing scene classification. MGCN extracts features from a feature extractor and reduces feature dispersion by an MGCN and DA to enhance the classification performance.

this indicator can be expressed as

$$\alpha_c = S_c / \hat{d}_c^2 \tag{3}$$

where $\hat{d}_c$ is defined as

$$\hat{d}_c = \min_{k \in C \wedge k \neq c} \left|\left| E\left[f\left(x^{(c)}\right)\right] - E\left[f\left(x^{(k)}\right)\right] \right|\right|_2. \tag{4}$$

As in [48], the intracategory variance for category c can be expressed as

$$S_c = \left|\left| \frac{1}{n_c - 1} \left(\boldsymbol{D}_c^{\mathrm{T}} \boldsymbol{D}_c - \frac{1}{n_c} \left(1^{\mathrm{T}} \boldsymbol{D}_c\right)^{\mathrm{T}} \left(1^{\mathrm{T}} \boldsymbol{D}_c\right) \right) \right|\right|_F \tag{5}$$

where $\boldsymbol{D}_c = [\, f(x_1^{(c)}), \quad f(x_2^{(c)}), \quad \ldots \quad f(x_{n_c}^{(c)})\,]^{\mathrm{T}}$ is composed of n_c feature vectors of samples from category c, $f(\cdot)$ denotes the feature extractor, $\boldsymbol{x}_i^{(c)}, i = 1, 2, \ldots, n_c$ denotes the ith sample, n_c refers to the number of samples in category c, $\mathbf{1}$ is an all-1 vector, and $\|\cdot\|_F$ denotes the Frobenius norm. The more similar the sample features of category c, the smaller the S_c is. In particular, if all the features are equal, $S_c = 0$. We use α to measure the average feature dispersion degree for all the N categories

$$\alpha = \frac{1}{N} \sum_{i=1}^{N} \alpha_i. \tag{6}$$

The smaller the α, the higher classification accuracy we can obtain. Through analysis and experiments, we find two methods can be used to reduce α: GCN and DA. GCN models the relationship of images by graph, making the features of samples from the same category more similar by learning on the graph and, consequently, reducing α. While DA can also reduce α, through training on D_A, the model can obtain high accuracy on D_A, which means that the categories from D_A can obtain a sufficiently small α. When DA is used to reduce the feature domain discrepancy between D_A and D_T, we can align these two distributions so that the common categories in D_T can obtain a comparatively smaller α and, thus, improve the classification accuracy. Based on the above analysis, we propose a few-shot classification method based on MGCN and an improved DA for the open set classification task.

B. Proposed MGCN

An overview of the proposed MGCN is presented in Fig. 4. The model consists of three main components: 1) a feature extractor, which is used to extract features from images; 2) an MGCN, which is used to model the relationships among images' features and use graph convolution to reduce the dispersion degree of category features; and 3) a metric prediction module, which predicts based on the metric distance in the feature space.

In the training phase, the pretrained ("pretrained" denotes the first two stages described in Section III-F) network is first used to perform a coarse classification on the unlabeled samples in D_T to estimate the pseudolabels. Then, DA is used to align the feature distributions between D_A and D_T with the samples from common categories. Since the classification accuracy of each category is approximately inversely related to α_c, in the training phase, we assign large coefficients to the domain loss functions for categories with large α_c and small coefficients to those with small α_c to reduce the average category dispersion degree. In addition, a graph is introduced to model the relationships among images' features, and the average feature dispersion degree is further reduced through graph convolution. Notably, we use the multiorder graph convolution instead of the first-order operation, which shows a more effective smoothing ability. In the testing phase, $N \times K$ labeled samples and Q unlabeled samples are fed into the network at the same time, and then, the predicted labels of the unlabeled samples are output.

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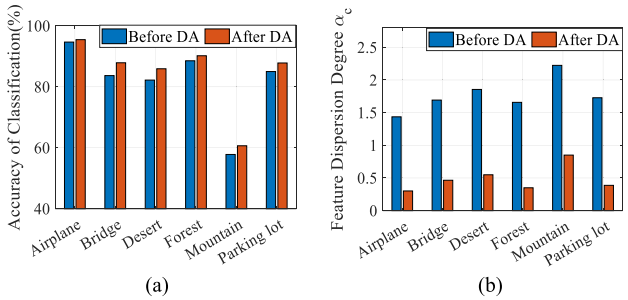


Fig. 5. Comparison of the classification accuracy and category feature dispersion α_c of ProtoNet before and after DA when the shot is 5, the training dataset is the common categories in RSI-CB256, the test dataset is the common categories in OPTIMAL-31, and the experimental setup is the same as in Section IV-C. (a) Classification accuracy. (b) Category feature dispersion α_c.

In Sections III-C and III-D, we will describe the proposed dispersion degree WDA and multiorder graph convolution in detail.

C. Dispersion Degree Weighting Domain Adaptation

Open set few-shot classification refers to that the label sets of D_A and D_T intersect partially. Since the sources of D_A and D_T are usually different, a model trained on D_A often cannot achieve sufficiently high classification accuracy for D_T due to the distribution discrepancy. Therefore, we introduce DA to reduce the domain discrepancy between auxiliary data and task data. Long et al. [49] and Wang et al. [33], [36] used formula (7) to calculate the interdomain discrepancy between D_A and D_T with the common labels

$$L_D = (1-\mu)||E[\phi(f(x_A))] - E[\phi(f(x_T))]||^2_{\mathcal{H}_k} + \mu \sum_{c=1}^{C} \left|\left|E\left[\phi\left(f\left(x_A^{(c)}\right)\right)\right] - E\left[\phi\left(f\left(x_T^{(c)}\right)\right)\right]\right|\right|^2_{\mathcal{H}_k} \tag{7}$$

where C is the number of the common categories, μ is a constant, $f(\cdot)$ denotes the feature extractor, $\mathcal{H}_k$ is a reproducing kernel Hilbert space (RKHS) [36], and $\phi(\cdot)$ defines a mapping from features to an RKHS. In RKHS, the inner product is defined as a kernel function $k(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle$, $k(x_i, x_j) = \exp(-\|x_i - x_j\|_2^2/2\sigma^2)$, and σ is a constant. The superscript c indicates that the sample belongs to the cth category.

In formula (7), L_D is defined by two terms. The first term reflects the overall feature distribution discrepancy between two datasets, which is called the marginal distribution discrepancy; the second term reflects the feature distribution discrepancy between the corresponding categories of the two datasets, which is called the conditional distribution discrepancy.

Fig. 5 shows the classification accuracy and α_c of ProtoNet before and after DA using formula (7). We find that DA can reduce the feature dispersion of samples in the same category. Intuitively, the larger the α_c, the more difficult it is to classify samples of the cth category. Therefore, to improve the effectiveness of DA and focus on the categories that are difficult to be classify, large coefficients should be assigned to the domain loss functions for categories with large α_c, and vice versa, during DA training. The final domain loss function is modified as follows:

$$L_D = (1-\mu)||E[\phi(f(x_A))] - E[\phi(f(x_T))]||^2_{\mathcal{H}_k} + \mu \sum_{c=1}^{C} \lambda_c \left|\left|E\left[\phi\left(f\left(x_A^{(c)}\right)\right)\right] - E\left[\phi\left(f\left(x_T^{(c)}\right)\right)\right]\right|\right|^2_{\mathcal{H}_k} \tag{8}$$

where $\lambda_c = Ce^{\alpha_c}/\sum_{k=1}^{C} e^{\alpha_k}$. In the training phase, L_D is minimized to reduce the dispersion degree of common categories and improve classification accuracy. $\mu \in [0, 1]$ is a parameter that balances between the marginal distribution discrepancy and the conditional distribution discrepancy. μ can dynamically adjust the importance of each distribution according to the actual data distribution. The value of μ reflects the contribution of marginal distribution discrepancy to the domain discrepancy; smaller μ corresponds to more contribution. The best classification results are obtained when μ is set to 0.1 by experiments. Therefore, μ is fixed at 0.1.

D. Multiorder Graph Convolution

A GCN is a deep learning network based on a graph. We first define a graph as follows: a graph is a data structure consisting of nodes and edges that are generally used to describe one-to-one or one-to-many relationships in a non-Euclidean space. A graph can be represented as $G = (\boldsymbol{V}, \boldsymbol{E}, \boldsymbol{A})$, where $\boldsymbol{V}$ is a set of nodes, E is a set of edges, $\boldsymbol{A} \in \mathbb{R}^{M\times M}$ is the adjacency matrix, $v_i \in \boldsymbol{V}$ denotes a node in the graph, and M denotes the number of nodes. $e_{ij} = (v_i, v_j) \in \boldsymbol{E}$, $i, j \in \{1, \ldots, M\}$, denotes an edge on the graph if $e_{ij} \in \boldsymbol{E}$ and $\boldsymbol{A}_{ij} > 0$; otherwise, $\boldsymbol{A}_{ij} = 0$.

In 2017, Kipf and Welling [45] proposed a concise and efficient graph convolution based on previous work

$$\boldsymbol{H}^{(l+1)} = \sigma\left(\tilde{\boldsymbol{A}}\boldsymbol{H}^{(l)}\boldsymbol{W}^{(l)}\right). \tag{9}$$

This method is commonly used in semisupervised classification tasks of graph nodes. $\tilde{\boldsymbol{A}} = \hat{\boldsymbol{D}}^{-1/2}\hat{\boldsymbol{A}}\hat{\boldsymbol{D}}^{-1/2}$ is a normalized augmented adjacency matrix, $\hat{\boldsymbol{A}} = \boldsymbol{A} + \boldsymbol{I}_N$ is an adjacency matrix of the undirected graph adding a self-loop, $\boldsymbol{D}_{ii} = \sum_j \hat{\boldsymbol{A}}_{ij}$ is the degree matrix, l denotes the number of layers in GCNs, $\boldsymbol{W}^{(l)}$ is a trainable weight, $\sigma(\cdot)$ is the activation function, $\boldsymbol{H}^{(l)} \in \mathbb{R}^{M\times D}$ is a matrix consisting of the feature vectors of M nodes in the lth layer of GCNs, and D is the dimension of the feature. The process of graph convolution is that each node on the graph aggregates the features of its neighbors and updates its own features given the inputs $\boldsymbol{A}$ and $\boldsymbol{H}^{(l)}$.

Recently, based on graph spectra analysis, it is noted that the multiplication of $\tilde{\boldsymbol{A}}$ by the feature matrix $\boldsymbol{H}^{(l)}$ for nodes in the graph convolution is equivalent to the low-pass filtering of $\boldsymbol{H}^{(l)}$ in the frequency domain [30]. The effect of low-pass filtering is to smooth node features in the graph, and nodes with the same labels will have similar features; thus, graph convolution can be used to reduce category dispersion degree, provide a more effective feature representation for metric prediction, and further improve classification accuracy.

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We construct a GCN using one batch of images from a dataset at a time. Node v_i, where $i \in \{1, \ldots, M\}$, represents an image, and M is the number of nodes. The number of labeled nodes in the GCN is $N \times K$, where N is the number of categories in D_T, K is the number of labeled nodes in each category, and the initial features of the nodes are extracted by the pretrained CNN. As in the literature [50], the radial basis function (RBF) is used to calculate the adjacency matrix

$$\boldsymbol{A}_{ij} = \begin{cases} e^{-\gamma ||\boldsymbol{h}_i - \boldsymbol{h}_j||_2^2}, & \text{if } j \in \mathcal{N}(i) \text{ and } i \in \mathcal{N}(j) \\ 0, & \text{else} \end{cases} \tag{10}$$

where $\boldsymbol{h}_i$ is the current feature vector of node i and $\mathcal{N}(i)$ refers to the KNNs of node i. In our experiments, $K = \lfloor M/N \rfloor - 1$, and $\gamma = 0.2$. Based on the above method, in most cases, a large weight is assigned between the nodes of the same category, and a small or zero weight is set between the nodes of different categories; consequently, in subsequent graph convolution, features associated with nodes of the same category are generally similar, and features associated with nodes of different categories have little influence on each other. This approach smooths the features of nodes from the same category and, thus, improves the classification accuracy of the model.

The experimental results in [45] and [51] show that proper smoothing can improve classification accuracy, and over-smoothing in graph convolution will deteriorate classification results. We find that the smoothing ability of the graph convolution is related to the orders of the adjacency matrix, so we propose a multiorder graph convolution

$$\boldsymbol{H}^{(l+1)} = \sigma\left(\boldsymbol{S}\boldsymbol{H}^{(l)}\boldsymbol{W}^{(l)}\right) \tag{11}$$

where $\boldsymbol{S} = \sum_{j=0}^{P} \beta_j \tilde{\boldsymbol{A}}^j$, β_j are weights that can be learned to adjust the smoothing ability, $\beta_j \geq 0$ and $\sum_{j=1}^{P} \beta_j = 1$, P denotes the highest order of the adjacency matrix, $\tilde{\boldsymbol{A}}$ is a normalized augmented adjacency matrix as in formula (9), and $\tilde{\boldsymbol{A}}^j$ is the jth power of $\tilde{\boldsymbol{A}}$. l denotes the number of layers; in our experiments, $l = 1$. The multiorder graph convolution is capable of adaptively adjusting the smoothing ability, and we analyze its smoothing ability in Appendix.

E. Metric Module

As in ProtoNet, the features output from a GCN are sent to the metric module to compute the category probabilities for the unlabeled nodes. The prototype center of each category is calculated with the labeled nodes by the following equation:

$$\boldsymbol{c}_k = \frac{1}{|S(k)|} \sum_{(\boldsymbol{x}_i, y_i) \in S(k)} \boldsymbol{h}_i, \quad k = 1, \ldots N \tag{12}$$

where $\boldsymbol{h}_i$ and y_i denote the feature vector and the label of node v_i, respectively. Then, the probability of each category is calculated by substituting the above result into a softmax function as follows:

$$p(y_i = k|\boldsymbol{h}_i) = \frac{\exp(-\|\boldsymbol{h}_i - \boldsymbol{c}_k\|_2)}{\sum_j \exp\left(-\|\boldsymbol{h}_i - \boldsymbol{c}_j\|_2\right)}. \tag{13}$$

F. Loss Function

The training of the MGCN can be divided into three stages: the pretraining stage of the feature extractor, the pretraining stage of the MGCN, and the DA stage. In the first stage, the feature extractor is pretrained on D_A to obtain initial parameters. In the second stage, the initial parameters of the feature extractor are loaded, and the whole model is trained on D_A to obtain the parameters of MGCN. At this point, MGCN is able to classify D_T. When there are overlapping categories between D_A and D_T, we use DA to further improve the classification accuracy. In the third stage, MGCN trained in the second stage is used to perform coarse classification on D_T, and the predicted label is used as the pseudolabel of the sample. The samples with a high probability of being predicted as common categories are selected for DA with the common category data in D_A. The loss functions used in the first and second stages are both categorical loss functions as in formula (14), and the loss function used in the third stage is the sum of categorical loss and domain loss as in formula (16)

$$L_A = -\sum_{k=1}^{n_A} y_A^k \log \hat{y}_A^k \tag{14}$$

$$L_T = -\sum_{k=1}^{n_T} \tilde{y}_T^k \log \hat{y}_T^k \tag{15}$$

$$L = \lambda_A L_A + \lambda_T L_T + \lambda_D L_D \tag{16}$$

where y_A^k denotes the true labels of D_A, $\tilde{y}_T$ denotes the pseudolabels of D_T, and $\hat{y}_A^k$ and $\hat{y}_T^k$ denote the predicted labels. λ_A, λ_T, and λ_D are three hyperparameters. To ensure that the auxiliary dataset and the task dataset make similar effects on the model, we take $\lambda_A = n_T/(n_T + n_A)$ and $\lambda_T = n_A/(n_T + n_A)$, where n_A and n_T are the numbers of samples from D_A and D_T, respectively. λ_D is the weight of the domain loss and is taken as 1 in the experiment. For a clearer explanation, we provide Algorithm 1 to describe the DA procedure in detail.[1]

IV. Experiments and Analysis

In this section, we first introduce the datasets used in the experiment in Section IV-A. Then, we describe the network architecture and the parameter settings in Section IV-B. Finally, we present and analyze the experimental results in Sections IV-C–IV-F. Among the results, Section IV-C shows the performance of the proposed method on three public datasets; Section IV-D studies the performance of the collected Tibetan Plateau dataset; and Sections IV-E and IV-F illustrate the influence of parameter P in the formula (11) and the depths of backbones on the performance, respectively.

A. Datasets

In the experiments, we evaluate the proposed MGCN on three publicly available datasets: AID [52], OPTIMAL-31 [53], RSI-CB256 [54], and a Tibetan Plateau dataset[2] that we

[1]The code will be available at https://github.com/chenjiehu/MGCN_ODA

[2]The collected Tibetan Plateau dataset is downloadable at https://drive.google.com/file/d/1q3U0xnYmrXvEdgBi8BuOArhMmXtNSBuJ/view?usp=sharing

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Algorithm 1 DA Procedure of the MGCN

Input: auxiliary dataset D_A and task dataset D_T, pre-trained parameters of the MGCN, probability threshold η.

Predict unlabeled samples with the MGCN and obtain pseudo labels, then select samples whose prediction probability is greater than η to form set D'_T.
for $i = 1$: max_iter do:
  Randomly sample auxiliary support set S_A and query set Q_A from D_A, sample task query set Q_T from D'_T;
  Predict the labels of samples in Q_A, and compute classification loss L_A using (14);
  Predict the labels of samples in Q_T, and compute classification loss L_T using (15);
  Compute domain loss L_D using (8) by common category samples in Q_A and Q_T;
  Compute total loss L using (16), update all the parameters in MGCN by back propagation;
end for
Test the MGCN on D_T.

Output: Average classification accuracy and classification results of D_T.

TABLE I
CHARACTERISTICS OF DIFFERENT DATASETS

Dataset	Category number	Sample number	Size	Pixel resolution/m
AID	30	10,000	600×600	0.5-8
OPTIMAL-31	31	1860	256×256	--
RSI-CB256	35	24,000	256×256	0.22-3
Tibetan Plateau	7	7,000	256×256	0.6-3

collected. They are listed in Table I. The above four datasets are acquired over different regions with different sensors, thus having distribution discrepancy across them [52], [53], [54].

To construct the open set classification task using publicly available datasets, in Section IV-C, we first randomly select two datasets D_A and D_T. This leads to six open set classification tasks: A→O, O→A, A→R, R→A, O→R, and R→O, where "A," "O," and "R" are abbreviations for AID, OPTIMAL-31, and RSI-CB256. The left-hand side of "→" indicates D_A, and the right-hand side indicates D_T. Second, following the common practice in few-shot classification [5], we split the training set, the validation set, and the test set, as shown in Tables II–IV (for example, for task A→O, the training and validation sets are from AID, and the test set is from OPTIMAL-31). The categories in bold are common categories.

B. Parameter Setting

We implement our model in the PyTorch framework, and the GPU is an NVIDIA GeForce RTX 3090. The batch size in the experiment is $N(K+Q)$, where N is the number of categories, K is the number of labeled samples in each category, Q is the number of samples to be predicted in each category, and $Q = 15$ in the experiment. The weight matrix $\boldsymbol{W} \in \mathbb{R}^{1600\times1000}$ in the MGCN is obtained through training. The outputs of the GCN are input into the batch normalization layer and the ReLU activation layer. Based on ReLU output, the final prediction is obtained by the metric module. The model is trained using the Adam optimizer; the initial learning rate is 0.001 and decreases by half every 50 epochs.

TABLE II
CATEGORIES OF TRAINING SET, VALIDATION SET, AND TEST SET WHEN TASK IS "A→O" OR "O→A"

Datasets	Training set (A→O) or test set (O→A)	Validation set (A→O)
AID	**Airport; Baseball field; Beach; Bridge; Church; Desert; Forest; Mountain; Parking;** Bare land; Center; Commercial; Dense residential; Farmland; Industrial; Meadow; Medium residential; Park; Playground; Pond;	Port; Railway station; Resort; River; School; Sparse residential; Square; Stadium; Storage tanks; Viaduct;
Datasets	**Training set (O→A) or test set (A→O)**	**Validation set (O→A)**
OPTIMAL-31	**Airport; Baseball diamond; Beach; Bridge; Church; Desert; Forest; Mountain; Parking lot;** Airplane; Basketball court; Chaparral; Circular farmland; Commercial area; Dense residential; Freeway; Golf course; Ground track field; Harbor; Industrial area;	Intersection; Island; Lake; Meadow; Medium residential; Mobile house park; Overpass; Railway; Rectangular farmland; Roundabout;

TABLE III
CATEGORIES OF TRAINING SET, VALIDATION SET, AND TEST SET WHEN TASK IS "A→R" OR "R→A"

Datasets	Training (A→R) or test set (R→A)	Validation set (A→R)
AID	**Bare land; Bridge; Desert; Forest; Mountain; Parking; River;** Baseball field; Beach; Center;	Airport; Church; Commercial; Dense residential; Farmland; Industrial; Meadow; Medium residential; Park; Playground;
Datasets	**Training set (R→A) or test set (A→R)**	**Validation set (R→A)**
RSI-CB256	**Bare land; Bridge; Desert; Forest; Mountain; Parking; River;** Baseball field; Beach; Center;	Avenue; City building; Coastline; Container; Crossroads; Dam; Dry farm; Green farmland; Highway; Pipeline;

C. Results and Comparisons on Public Datasets

To evaluate the performance of the proposed method, in this section, we adopt five existing few-shot learning methods for experimental comparisons: MAML [20], MAML++ [55],

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TABLE IV
CATEGORIES OF TRAINING SET, VALIDATION SET, AND TEST SET WHEN TASK IS "O→R" OR "R→O"

Datasets	Training set (O→R) or test set (R→O)	Validation set (O→R)
OPTIMAL -31	**Airplane; Bridge; Desert; Forest; Mountain; Parking lot;** Airport; Baseball field; Basketball court; Beach; Chaparral; Church;	Circular farmland; Commercial area; Dense residential; Freeway; Golf course; Ground track field; Harbor; Industrial area; Intersection;
Datasets	Training set (R→O) or test set (O→R)	Validation set (R→O)
RSI-CB256	**Airplane; Bridge; Desert; Forest; Mountain; Parking lot;** Airport runway; Artificial grassland; Avenue; Bare land; City building; Coastline;	Container; Crossroads; Dam; Dry farm; Green farmland; Highway; Hirst; Lakeshore; Mangrove; Marina; Pipeline;

TABLE V
COMPARISON OF DIFFERENT METHODS

Method	Feature Extractor	Adjacency Matrix Calculation	Is feature learning performed?	Label Prediction
MAML	Conv4	No	No	Linear Classifier
MAML++	Conv4	No	No	Linear Classifier
ProtoNet	Conv4	No	No	Metric Distance
Proto-MAML	Conv4	No	No	Metric Distance
TPN	Conv4	KNN+ RBF	No	Label Probability
MGCN	Conv4	KNN+ RBF	Yes	Metric Distance

Proto-MAML [56], ProtoNet [8], and TPN [28]. The key points of the proposed method and the compared methods are shown in Table V, where Conv4 denotes four-layer convolutional networks [8].

We evaluate the experimental results using classification accuracy, which is defined as

$$\text{Acc} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}. \tag{17}$$

The classification accuracies of the proposed method and the compared methods are shown in Tables VI–XI for different tasks. "no DA" means that the method is tested directly on D_T without DA, "DA" indicates that the method is trained with the DA method in [37], "WDA" means that the method is trained with the proposed DA approach, and "TPN + GCN" means that TPN is combined with the proposed MGCN without DA.

In addition, to investigate the effect of both DA and multiorder graph convolution on the common categories and private categories, the results show three kinds of classification accuracies: average accuracy for common categories Acc_c, average accuracy for private categories Acc_p, and average accuracy for all the categories Acc. The experimental results are computed averagely on 20 randomly generated episodes from the test set.

From Tables VI to XI, it can be concluded as follows.

1) Comparing ProtoNet with MGCN or TPN with TPN + GCN, Acc_c, Acc_p, and Acc are improved by the proposed multiorder graph convolution in most instances. This reveals that the proposed multiorder graph convolution can significantly improve the accuracy for both common categories and private categories.

2) Comparing accuracy of the methods before and after DA, Acc_c, and Acc of all the methods improve markedly after DA, while Acc_p does not increase significantly, which indicates that DA can significantly improve the accuracy for the common categories.

3) Comparing "DA" with "WDA," it can be concluded that the proposed DA is better than the DA method in [37]; it further improves the classification accuracy by approximately 1%.

4) The proposed MGCN_{WDA} achieves the best result in most cases and significantly outperforms the "no DA, no GCN" methods, i.e., MAML, MAML++, and Proto-MAML.

Table XII shows the average category feature dispersion degree α of the five methods. From the table, it can be concluded that both DA and graph convolution can reduce the feature dispersion degree α, and the smaller the α, the more accurate the classification results are. DA and GCNs can improve the classification accuracy of ProtoNet because they reduce the dispersion degree of those common category samples in the feature space.

Although the proposed method is designed by combining ProtoNet with the proposed MGCN and WDA, it can be seen from Tables VI to XI that the proposed MGCN and WDA can be integrated with other few-shot classification or open set few-shot classification methods (such as TPN [28]) to reduce the degree of feature dispersion and improve the classification accuracy.

D. Results and Comparisons on Tibetan Plateau Dataset

The ultimate purpose of this work is to apply the few-shot classification to a Tibetan Plateau scientific research project. In this section, an experiment is conducted based on the collected Tibetan Plateau dataset using the proposed MGCN, ProtoNet, and TPN. The Tibetan Plateau dataset has seven categories: bare land, farmland, forest, river, building, grassland, and snow mountain. In the experiment, the training set is selected from AID, including bare land, farmland, forest, river, airport, baseball field, beach, bridge, center, church, commercial, dense residential, desert, industrial, meadow, medium residential, mountain, park, parking, and playground, a total of 20 categories. Other ten categories in AID are selected as the validation set. There are four categories that are the same in both datasets: bare land, farmland, forest, and river, as shown in Fig. 6.

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TABLE VI

CLASSIFICATION ACCURACIES (%) FOR COMMON CATEGORIES, PRIVATE CATEGORIES, AND ALL CATEGORIES WHEN THE TASK IS A→O

	20-way 1-shot			20-way 5-shot		
	Acc_c	Acc_p	Acc	Acc_c	Acc_p	Acc
MAML	32.69±1.14	32.60±1.21	32.64±0.97	49.39±0.91	49.18±0.87	49.27±0.86
MAML++	47.36±0.44	32.49±0.52	39.18±0.35	65.38±0.39	43.01±0.46	53.07±0.31
Proto-MAML	38.76±0.41	30.36±0.48	34.14±0.38	67.55±0.35	49.99±0.33	57.89±0.29
ProtoNet						
no DA	35.43±0.40	24.36±0.48	29.34±0.26	67.94±0.23	47.70±0.25	56.81±0.19
DA	39.09±0.61	24.22±0.56	30.91±0.44	73.43±0.38	**51.07±0.54**	61.13±0.35
TPN						
no DA	49.15±0.37	30.66±0.43	38.98±0.27	68.08±0.23	46.72±0.49	56.33±0.24
DA	50.79±0.44	32.52±0.47	40.74±0.36	75.57±0.36	50.17±0.48	61.60±0.32
TPN+GCN	45.38±0.72	**34.82±0.66**	39.57±0.62	71.41±0.51	49.77±0.63	59.51±0.47
MGCN						
no DA	45.83±0.79	32.71±0.68	38.61±0.56	69.11±0.55	49.06±0.42	58.08±0.38
DA	50.66±0.81	32.99±0.73	40.94±0.69	75.89±0.69	48.85±0.61	61.02±0.59
WDA	**51.01±0.89**	33.45±0.78	**41.35±0.77**	**77.98±0.70**	49.90±0.57	**62.53±0.53**

TABLE VII

CLASSIFICATION ACCURACIES (%) FOR COMMON CATEGORIES, PRIVATE CATEGORIES, AND ALL CATEGORIES WHEN THE TASK IS O→A

	20-way 1-shot			20-way 5-shot		
	Acc_c	Acc_p	Acc	Acc_c	Acc_p	Acc
MAML	28.44±1.51	28.38±1.26	28.41±1.23	39.82±1.07	40.01±1.54	39.93±1.14
MAML++	33.71±0.85	28.45±0.73	30.81±0.72	42.60±0.74	45.05±0.79	43.95±0.69
Proto-MAML	36.30±0.77	25.79±0.82	30.52±0.64	60.35±0.65	46.20±0.68	52.57±0.57
ProtoNet						
no DA	27.26±0.61	22.37±0.80	24.57±0.57	57.84±0.49	45.09±0.65	50.83±0.48
DA	29.33±0.89	24.08±0.91	26.44±0.77	55.80±0.52	49.84±0.60	52.52±0.41
TPN						
no DA	43.55±0.75	28.47±0.77	35.26±0.62	61.38±0.78	49.03±0.70	54.59±0.63
DA	46.31±0.86	28.64±0.91	36.59±0.78	**67.73±0.84**	54.56±0.89	60.49±0.75
TPN+GCN	41.33±0.79	30.09±0.65	35.15±0.61	63.04±0.72	52.75±0.81	57.38±0.68
MGCN						
no DA	40.65±1.17	**31.59±0.75**	35.67±0.78	63.48±0.96	52.92±0.80	57.67±0.69
DA	47.05±1.37	28.82±1.12	37.02±0.92	65.13±0.86	**56.11±0.94**	60.17±0.85
WDA	**47.49±1.25**	29.33±1.20	**37.50±0.94**	66.64±1.03	56.04±0.82	**60.81±0.89**

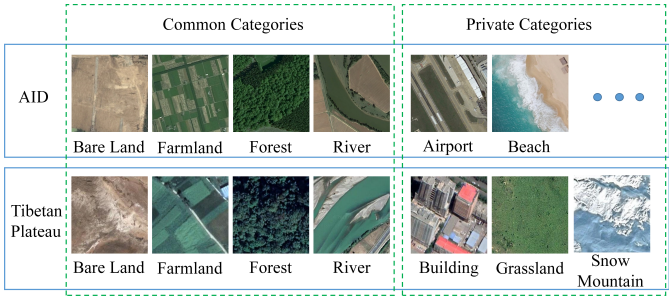


Fig. 6. Illustration of the AID and Tibetan Plateau dataset. There are four common categories and three private categories in the Tibetan Plateau dataset.

The classification accuracies of different methods on the Tibetan Plateau dataset are given in Tables XIII and XIV when the labeled samples of each category are 1 and 5, respectively.

The confusion matrices when the shot is 5 are presented in Fig. 7. From Fig. 7 and Tables XIII and XIV, similar to Section IV-C, the proposed method achieves good performance on the Tibetan Plateau dataset.

Fig. 8 shows those α_c corresponding to different categories. Comparing $\text{ProtoNet}_{\text{noDA}}$ and $\text{MGCN}_{\text{noDA}}$, we can see that α_c is reduced after adding the GCN part, indicating that GCN can effectively reduce the dispersion of the category features. Comparing $\text{MGCN}_{\text{noDA}}$, MGCN_{DA}, and MGCN_{WDA}, it can be seen that, after using DA, the dispersion of categories is reduced for both common categories and private categories.

Comparing MGCN_{WDA} with MGCN_{DA}, it can be seen that, for the common categories, MGCN_{WDA} has lower α_c than that of MGCN_{DA} on bare land, farmland, and river, and slightly higher α_c than that of MGCN_{DA} on the forest. This is because α_c of the forest before DA is already very low, which indicates

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TABLE VIII

CLASSIFICATION ACCURACIES (%) FOR COMMON CATEGORIES, PRIVATE CATEGORIES, AND ALL CATEGORIES WHEN THE TASK IS A→R

	10-way 1-shot			10-way 5-shot		
	Acc_c	Acc_p	Acc	Acc_c	Acc_p	Acc
MAML	41.43±0.99	40.81±0.94	41.25±0.86	61.84±0.91	63.21±0.87	62.25±0.78
MAML++	45.55±0.54	44.94±0.50	45.37±0.41	64.11±0.48	46.72±0.51	58.89±0.40
Proto-MAML	37.16±0.41	41.16±0.47	38.36±0.39	68.12±0.34	59.15±0.62	65.43±0.36
ProtoNet						
no DA	33.42±0.25	44.13±0.56	36.63±0.32	67.50±0.27	58.91±0.48	64.92±0.30
DA	38.93±0.56	**47.06±0.73**	41.37±0.54	**75.31±0.55**	53.65±0.69	68.82±0.51
TPN						
no DA	48.43±0.40	34.10±0.42	44.13±0.27	61.13±0.38	66.24±0.47	62.66±0.23
DA	51.45±0.57	42.52±0.49	48.77±0.42	64.79±0.51	64.59±0.43	64.73±0.31
TPN+GCN	50.71±1.28	40.31±1.07	47.59±0.97	62.34±0.60	**66.94±0.52**	63.72±0.47
MGCN						
no DA	47.50±1.03	42.98±0.72	46.14±0.84	64.84±0.71	64.64±0.84	64.77±0.69
DA	51.66±1.32	43.56±0.95	49.23±1.02	72.52±0.85	59.75±0.93	68.69±0.79
WDA	**52.45±1.21**	46.22±0.90	**50.58±0.96**	73.19±1.04	60.96±0.98	**69.52±0.90**

TABLE IX

CLASSIFICATION ACCURACIES (%) FOR COMMON CATEGORIES, PRIVATE CATEGORIES, AND ALL CATEGORIES WHEN THE TASK IS R→A

	10-way 1-shot			10-way 5-shot		
	Acc_c	Acc_p	Acc	Acc_c	Acc_p	Acc
MAML	38.14±0.94	**38.53±0.82**	38.26±0.69	53.68±0.80	54.20±0.77	53.83±0.61
MAML++	48.77±0.47	15.20±0.40	38.70±0.29	54.51±0.32	49.36±0.39	52.97±0.26
Proto-MAML	42.39±0.49	19.92±0.37	35.65±0.32	54.19±0.38	61.19±0.29	56.29±0.27
ProtoNet						
no DA	39.71±0.42	14.88±0.53	32.26±0.25	51.52±0.21	62.79±0.40	54.90±0.17
DA	45.20±0.43	18.30±0.57	37.13±0.39	62.25±0.23	57.10±0.63	60.71±0.27
TPN						
no DA	49.13±0.38	24.38±0.29	41.71±0.24	57.02±0.20	62.32±0.23	58.61±0.17
DA	53.56±0.56	26.93±0.42	45.57±0.35	62.83±0.46	56.75±0.33	61.00±0.25
TPN+GCN	51.13±0.62	27.43±0.74	44.02±0.61	61.07±0.62	62.60±0.57	61.53±0.46
MGCN						
no DA	47.57±0.93	38.38±0.62	44.81±0.75	57.55±0.83	66.15±0.50	60.13±0.66
DA	56.74±1.19	29.74±0.71	48.64±0.72	62.05±0.71	**68.12±0.51**	63.87±0.49
WDA	**57.89±1.25**	28.36±0.84	**49.03±0.85**	**64.20±1.09**	67.20±0.72	**65.10±0.86**

that we are able to obtain high classification accuracy without DA. Thus, in the process of DA, we assign large weights to bare land, farmland, and river. Just as we expected, MGCN_{WDA} obtains higher average classification accuracy than MGCN_{DA}.

In particular, from Fig. 7, we can find that the proposed method still gets high accuracy on the three private categories of the forest, grassland, and snow mountain though they are no corresponding categories in the training data. From Fig. 8, we can see that the three private categories have small α_c, which means that they have small intercategory similarity and large intracategory similarity.

E. Effect of P on Accuracy

In order to investigate the influence of the adjacency matrix's order on accuracy in the multiorder graph convolution and eliminate the influence of DA, this experiment tests the classification accuracy of the proposed MGCN on three different datasets when the values of P in (12) are 1, 2, 3, 4, and 5, respectively, without using DA. The experimental setup is the same as in Section IV-C, and the results are shown in Fig. 9.

As can be seen from Fig. 9, with one or five labeled samples in each category, $P = 2$ is able to obtain a 0.5%–3% improvement compared to $P = 1$. According to the analysis in Appendix and the experiments in this section, increasing P enhances the smoothing ability of graph convolution. However, when P is greater than 3, there is a small increase in classification accuracy in some datasets and a small decrease in others. This may be because a larger P will produce oversmoothing to some extent, resulting in a loss of accuracy [29]. In addition, increasing P will significantly increase the computational complexity. Therefore, P is set to 2 in the experiment.

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TABLE X
CLASSIFICATION ACCURACIES (%) FOR COMMON CATEGORIES, PRIVATE CATEGORIES, AND ALL CATEGORIES WHEN THE TASK IS O→R

	12-way 1-shot			12-way 5-shot		
	Acc_c	Acc_p	Acc	Acc_c	Acc_p	Acc
MAML	39.21±1.71	38.71±1.59	38.96±1.39	57.31±1.52	56.91±1.43	57.11±1.22
MAML++	22.41±1.05	53.11±0.92	37.76±0.78	54.89±0.94	60.77±0.79	57.83±0.68
Proto-MAML	37.01±1.08	35.43±0.96	36.22±0.82	60.41±0.90	68.69±0.81	64.55±0.69
ProtoNet						
no DA	34.06±0.84	34.36±0.98	34.21±0.73	59.35±0.66	68.45±0.53	63.90±0.46
DA	20.32±1.08	51.02±0.88	35.67±0.96	63.60±1.04	69.48±0.82	66.54±0.74
TPN						
no DA	40.02±0.95	45.48±1.24	42.75±0.84	55.68±0.92	66.38±0.79	61.03±0.67
DA	42.78±1.09	53.00±1.46	47.89±1.05	66.34±0.96	64.52±1.11	65.43±0.80
TPN+GCN	41.59±0.83	48.47±1.25	45.03±0.85	60.11±0.78	66.59±1.05	63.35±0.71
MGCN						
no DA	44.05±1.87	50.57±1.44	47.31±1.57	61.72+0.87	**70.40±0.93**	66.06±0.82
DA	44.07±1.49	**53.81±1.67**	48.94±1.23	69.47±1.34	65.43±1.21	67.45±0.99
WDA	**45.43±1.56**	53.61±1.71	**49.52±1.43**	**70.70±1.21**	67.22±1.04	**68.96±0.85**

TABLE XI
CLASSIFICATION ACCURACIES (%) FOR COMMON CATEGORIES, PRIVATE CATEGORIES, AND ALL CATEGORIES WHEN THE TASK IS R→O

	12-way 1-shot			12-way 5-shot		
	Acc_c	Acc_p	Acc	Acc_c	Acc_p	Acc
MAML	37.46±0.94	34.48±1.10	35.97±0.86	48.85±1.02	48.23±0.79	48.54±0.75
MAML++	45.72±0.63	**37.60±0.84**	41.66±0.54	68.71±0.66	49.66±0.53	59.18±0.41
Proto-MAML	36.74±0.75	33.16±0.79	34.95±0.59	68.61±0.71	51.52±0.59	60.06±0.47
ProtoNet						
no DA	37.41±0.71	21.37±0.45	29.39±0.47	66.73±0.37	**52.79±0.42**	59.76±0.22
DA	43.69±0.81	37.47±1.01	40.58±0.83	75.06±0.80	50.01±0.94	62.53±0.81
TPN						
no DA	55.27±0.67	24.87±0.72	40.07±0.58	73.57±0.74	49.19±0.61	61.38±0.55
DA	59.02±1.17	26.82±1.08	42.92±0.91	75.16±0.50	49.42±0.44	62.29±0.36
TPN+GCN	58.40±0.71	24.08±0.92	41.24±0.62	74.01±0.83	50.15±0.77	62.08±0.68
MGCN						
no DA	56.88±0.83	24.71±0.97	40.80±0.77	75.40±0.51	48.52±0.66	61.96±0.42
DA	58.39±1.16	26.91±1.05	42.65±0.90	76.28±0.39	48.58±0.46	62.43±0.31
WDA	**59.16±1.20**	27.45±1.12	**43.31±0.93**	**76.74±0.97**	49.88±0.84	**63.31±0.77**

TABLE XII
CATEGORY FEATURE DISPERSION DEGREE α FOR EACH METHOD WHEN SHOT IS FIVE

Method	Datasets		
	A⟶O	A⟶R	O⟶R
ProtoNet			
no DA	1.8603	1.8049	2.6817
DA	1.3723	1.1544	1.8169
MGCN			
no DA	1.4676	1.2466	2.4509
DA	1.0680	1.0658	1.6587
WDA	1.0169	1.0598	1.6426

F. Influence of Different Backbones on Accuracy

To study the influence of different backbones on the performance of different methods, we compare four methods, MGCN, ProtoNet, Baseline++ [57], and Baseline++* [57], when the backbone networks are Conv4, Conv6, ResNet10, ResNet18, and ResNet34, respectively. Baseline++ and Baseline++* are two fine-tuning methods; they are pretrained on a public dataset and fine-tuned on the limited labeled samples from D_T. The difference is that Baseline++ is only pretrained on D_A(D_A is OPTMAL-31 or RSI-CB256), and Baseline++* is pretrained on both a public dataset Mini-Imagenet [8] and D_A, which aims at testing whether the accuracy can be improved after being pretrained on a larger public dataset. In this experiment, OPTIMAL-31 and RSI-CB256 are used as D_A and D_T, and the setup is the same as in Section IV-C. The results are the average of 20 runs ± standard deviation and are shown in Fig. 10.

The accuracy of MGCN is better than that of ProtoNet in most cases in both DA and noDA. Compared with $\text{ProtoNet}_{\text{DA}}$, MGCN_{WDA} adds a graph network, and the other parts are

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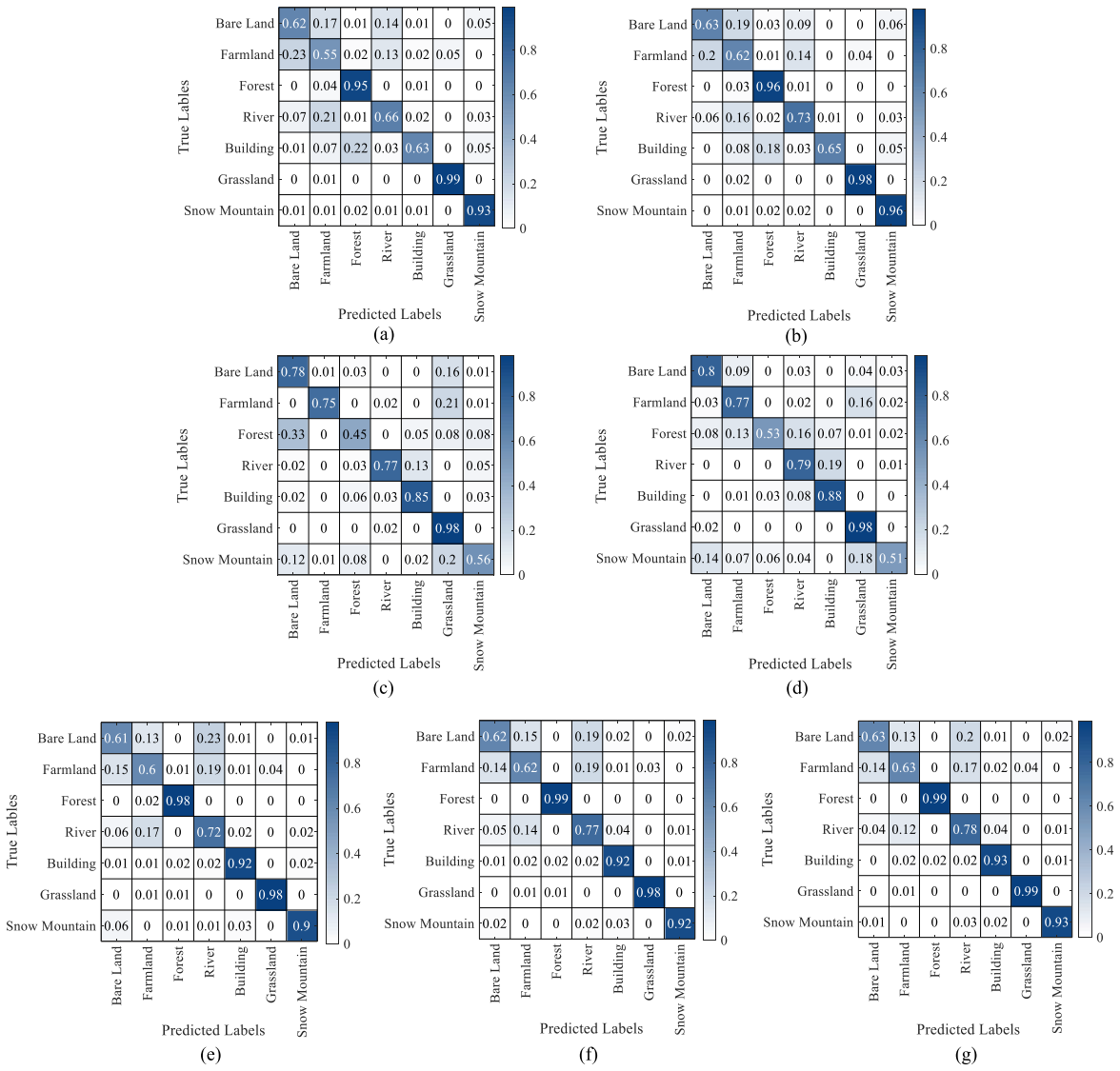


Fig. 7. Confusion matrix for seven methods when the shot is 5. The auxiliary dataset is AID, and the task dataset is the Tibetan Plateau. (a) $\text{ProtoNet}_{\text{noDA}}$. (b) $\text{ProtoNet}_{\text{DA}}$. (c) TPN_{noDA}. (d) TPN_{DA}. (e) $\text{MGCN}_{\text{noDA}}$. (f) MGCN_{DA}. (g) MGCN_{WDA}.

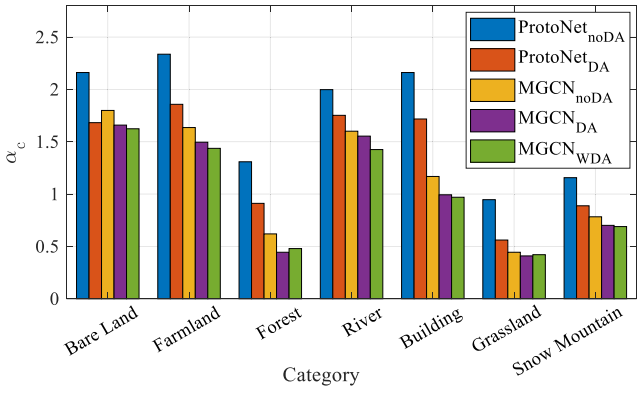


Fig. 8. Category feature dispersion α_c for each method trained on AID and tested on the Tibetan Plateau dataset when the shot is 5.

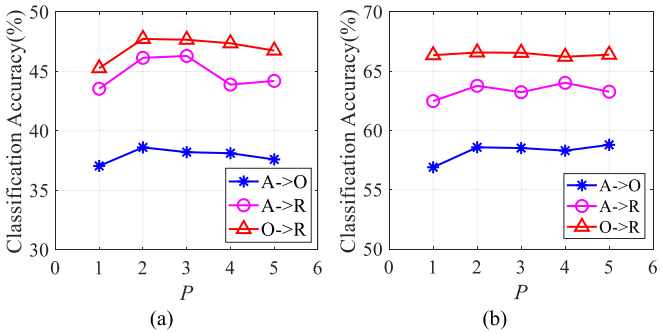


Fig. 9. Classification accuracy with different P values. (a) Shot is 1. (b) Shot is 2.

the same. We can conclude that, after adding the graph network, the classification accuracy can be improved since the relationships among samples are explicitly modeled, and this helps to smooth the intraclass features and reduce the intraclass feature dispersion.

In addition, the experimental results show that, when the backbone is ResNet10, most methods achieve the best results. As the backbones get deeper, the classification accuracy of

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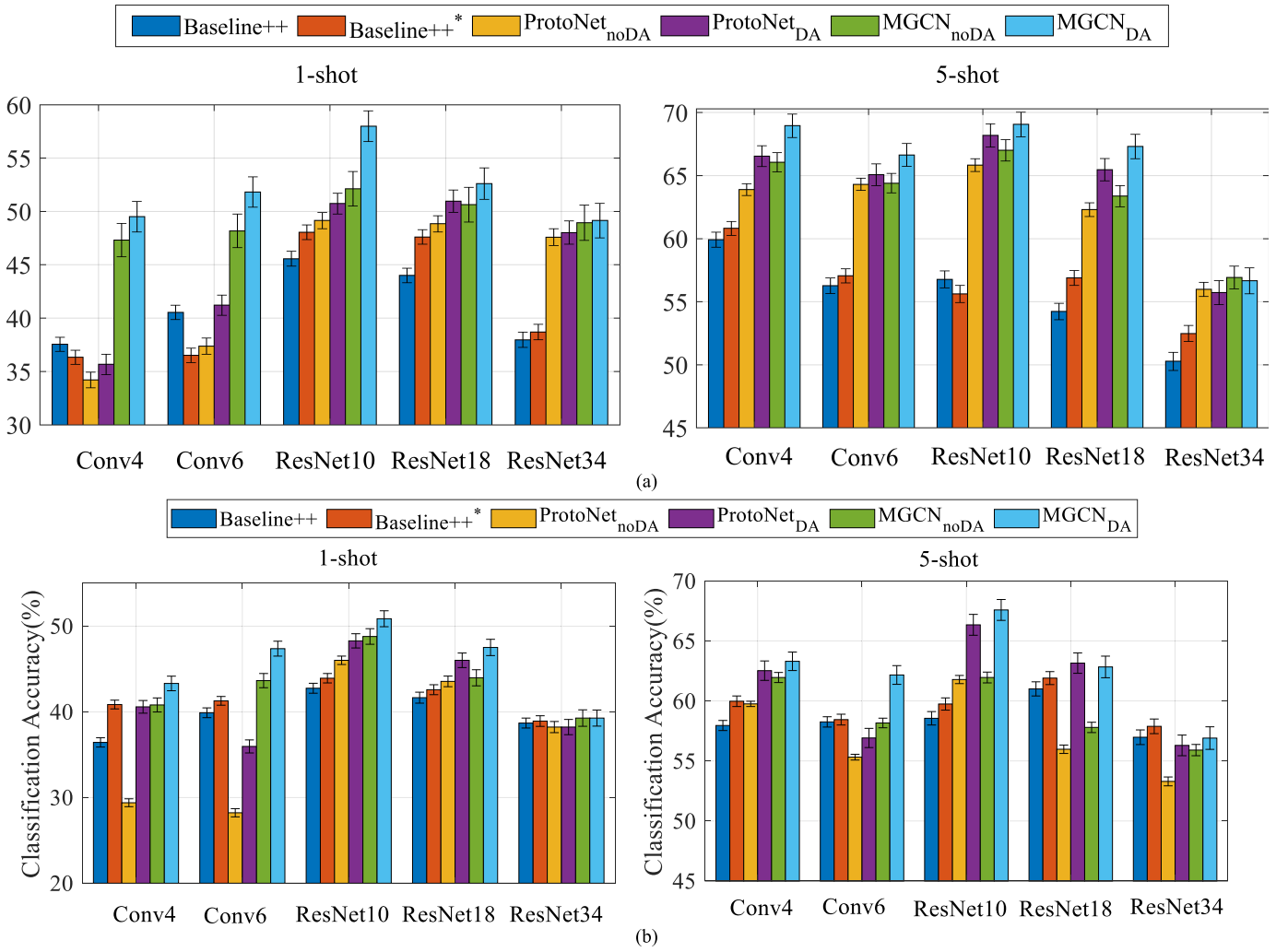


Fig. 10. Classification accuracy with standard deviation bars of different methods under different backbones when the datasets are (a) O→R and (b) R→O, respectively.

TABLE XIII

CLASSIFICATION ACCURACIES (%) FOR COMMON CATEGORIES, PRIVATE CATEGORIES, AND ALL CATEGORIES WHEN SHOT IS ONE

Method	7way-1shot		
	Acc_c	Acc_p	Acc
ProtoNet			
no DA	49.15±0.67	69.31±0.73	57.79±0.53
DA	51.93±0.94	70.47±0.96	59.87±0.79
TPN			
no DA	51.36±0.85	73.20±0.90	60.72±0.66
DA	56.02±1.12	72.49±0.93	63.08±0.84
MGCN			
no DA	55.77±1.05	78.56±0.97	65.54±0.80
DA	60.52±1.17	**79.67±1.32**	68.72±0.97
WDA	**63.07±1.26**	78.30±1.25	**69.60±1.02**

TABLE XIV

CLASSIFICATION ACCURACIES (%) FOR COMMON CATEGORIES, PRIVATE CATEGORIES, AND ALL CATEGORIES WHEN SHOT IS FIVE

Method	7way-5shot		
	Acc_c	Acc_p	Acc
ProtoNet			
no DA	69.67±0.59	84.78±0.63	76.14±0.44
DA	73.37±0.72	86.10±0.66	78.83±0.52
TPN			
no DA	68.74±0.68	79.78±0.69	73.47±0.52
DA	72.30±0.87	78.89±0.74	75.12±0.61
MGCN			
no DA	72.84±0.88	93.32±0.91	81.62±0.69
DA	74.91±0.95	94.38±1.08	83.26±0.82
WDA	**75.96±1.06**	**94.83±1.07**	**84.05±0.84**

most methods decreases to a certain extent. There are two possible reasons: one is that deeper backbones require a large amount of labeled data for training, but the remote sensing images for training are usually insufficient; the other is probably caused by the characteristics of remote sensing images. For remote sensing scenes, the key features used to distinguish among different categories are usually embedded in the details of the image, and the features extracted by deeper

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backbones tend to ignore the details of the image [58], [59]. Therefore, in remote sensing image few-shot classification, deeper backbones may lead to performance degradation.

In addition, Chen et al. [57] point out that using deeper networks and fine-tuning methods can also be effective in solving few-shot classification problems. In order to compare the proposed MGCN with such a simple setup, we carry out the experiments with Baseline++ and Baseline++* under different backbones and show the results in Fig. 10. From the results, we can get that the classification accuracy of the proposed method is still higher than those of Baseline++ and Baseline++*. This is because the fine-tuning methods cannot solve the problem of feature distribution discrepancy between two datasets and neglect the relations among samples. The above experiments and analysis indicate that the proposed MGCN outperforms the common simple setup strategies.

V. Conclusion and Future Work

To solve the problem that the category sets of auxiliary data and task data may partially intersect in few-shot classification tasks in remote sensing images, we propose an open set few-shot remote sensing scene classification method based on an MGCN and WDA. In the method, a graph is utilized to model the relationships among images, and the features of nodes are updated by multiorder graph convolution to reduce the intracategory feature dispersion for each category. In addition, we propose a WDA method to reduce the domain discrepancy between auxiliary data and task data. To verify the effectiveness of the proposed method, we compare the proposed method with ProtoNet, TPN, MAML, MAML++, and Proto-MAML with the same feature extraction networks. The experimental results show that the proposed method outperforms the compared methods.

In the future, we will extend the proposed multiorder graph convolution and WDA to other few-shot learning tasks, such as remote sensing image segmentation and change detection.

Appendix

In this section, we will analyze the smoothing ability of the graph convolution to demonstrate the effectiveness of the proposed multiorder graph convolution in Section III-D.

The graph signals' smoothness is generally measured using total variation [51], which is defined as follows:

$$\Delta(\boldsymbol{H}) = tr\left(\boldsymbol{H}^{\mathrm{T}}\tilde{\boldsymbol{L}}\boldsymbol{H}\right) = \sum_{\left(v_i,v_j\right)\in \boldsymbol{E}} \left\| \frac{\boldsymbol{h}_i}{\sqrt{d_i+1}} - \frac{\boldsymbol{h}_j}{\sqrt{d_j+1}} \right\|_2^2 \tag{A1}$$

where $\tilde{\boldsymbol{L}} = \boldsymbol{I} - \tilde{\boldsymbol{A}}$ is the normalized Laplace matrix, $\tilde{\boldsymbol{A}}$ is the same as in formula (9), $\boldsymbol{H}$ is the feature matrix, $d_i = \hat{\boldsymbol{D}}_{ii}$, and $\hat{\boldsymbol{D}}_{ii}$ is defined in Section III-D. The smaller $\Delta(\boldsymbol{H})$ is, the smoother the $\boldsymbol{H}$. Here, we show that the graph convolution in formula (9) can reduce $\Delta(\boldsymbol{H})$, and the proposed multiorder graph convolution in formula (11) can further adaptively enhance the smoothing capability of the graph convolution.

Li et al. [46] denote that the major smoothing function in GCNs is the multiplication of $\tilde{\boldsymbol{A}}$ and $\boldsymbol{H}$, which can be written as

$$\boldsymbol{H}_{\mathrm{out}} = \tilde{\boldsymbol{A}}\boldsymbol{H} \tag{A2}$$

where $\boldsymbol{H}_{\mathrm{out}}$ refers to the feature matrix after multiplying by $\tilde{\boldsymbol{A}}$. Thus, we have

$$\begin{aligned}
\Delta(\boldsymbol{H}_{\mathrm{out}}) &= \Delta\left(\tilde{\boldsymbol{A}}\boldsymbol{H}\right) \\
&= \mathrm{tr}\left(\boldsymbol{H}_{\mathrm{out}}^{\mathrm{T}}\tilde{\boldsymbol{L}}\boldsymbol{H}_{\mathrm{out}}\right) \\
&= \mathrm{tr}\left(\boldsymbol{H}_{\mathrm{out}}^{\mathrm{T}}\boldsymbol{U}^{T}\boldsymbol{\Lambda}\boldsymbol{U}\boldsymbol{H}_{\mathrm{out}}\right) \\
&= \mathrm{tr}\left(\boldsymbol{H}^{\mathrm{T}}\boldsymbol{A}^{\mathrm{T}}\boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{H}\right) \\
&= \mathrm{tr}\left(\boldsymbol{H}^{\mathrm{T}}\boldsymbol{U}(\boldsymbol{I}-\boldsymbol{\Lambda})\boldsymbol{U}^{\mathrm{T}}\boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^{\mathrm{T}}\boldsymbol{U}(\boldsymbol{I}-\Lambda)\boldsymbol{U}^{\mathrm{T}}\boldsymbol{H}\right) \\
&= \mathrm{tr}\left(\boldsymbol{H}^{\mathrm{T}}\boldsymbol{U}\boldsymbol{\Lambda}(\boldsymbol{I}-\boldsymbol{\Lambda})^2\boldsymbol{U}^{\mathrm{T}}\boldsymbol{H}\right) \\
&= \sum_{i=1,\ldots M} (1-\lambda_i)^2\lambda_i \left|\left|\left(\boldsymbol{U}^{\mathrm{T}}\boldsymbol{H}\right)_i\right|\right|_2^2
\end{aligned} \tag{A3}$$

where λ_i, $i = 1, 2, \ldots, M$, are the M eigenvalues of $\tilde{\boldsymbol{L}}$, $\boldsymbol{\Lambda}$ is a diagonal matrix composed of λ_i, $\boldsymbol{U}$ is the corresponding orthogonal matrix, and $(\boldsymbol{U}^{\mathrm{T}}\boldsymbol{H})_i$ denotes the ith row of $\boldsymbol{U}^{\mathrm{T}}\boldsymbol{H}$. Since $\lambda_i \in [0, 1.5]$ [46], we can obtain

$$\begin{aligned}
\Delta(\boldsymbol{H}_{\mathrm{out}}) &= \sum_{i=1,\ldots M} (1-\lambda_i)^2\lambda_i \left|\left|\left(\boldsymbol{U}^{\mathrm{T}}\boldsymbol{H}\right)_i\right|\right|_2^2 \\
&\le \sum_{i=1,\ldots M} \lambda_i \left|\left|\left(\boldsymbol{U}^{\mathrm{T}}\boldsymbol{H}\right)_i\right|\right|_2^2 = \Delta(\boldsymbol{H}).
\end{aligned} \tag{A4}$$

Therefore, the graph convolution can reduce $\Delta(\boldsymbol{H})$ and act as a smoothing effect on the signal on the graph.

For the graph convolution proposed in this article, there are $\boldsymbol{H}_{\mathrm{out}} = \boldsymbol{S}\boldsymbol{H}$; we can obtain

$$\begin{aligned}
\Delta(\boldsymbol{H}_{\mathrm{out}}) &= \Delta(\boldsymbol{S}\boldsymbol{H}) \\
&= \Delta\left(\sum_{j=0}^{P} \beta_j \tilde{\boldsymbol{A}}^j \boldsymbol{H}\right) \\
&= \sum_{i=1,\ldots M} \sum_{j=0,\ldots P} \beta_j (1-\lambda_i)^{2j} \left|\left|\lambda_i \left(\boldsymbol{U}^{\mathrm{T}}\boldsymbol{H}\right)_i\right|\right|_2^2.
\end{aligned} \tag{A5}$$

In addition, because $\beta_j \ge 0$ and $\sum_{j=1}^{P} \beta_j = 1$, we have

$$\begin{aligned}
&\sum_{i=1,\ldots M} (1-\lambda_i)^{2P} \left|\left|\lambda_i \left(\boldsymbol{U}^{\mathrm{T}}\boldsymbol{H}\right)_i\right|\right|_2^2 \\
&\quad\le \sum_{i=1,\ldots M} \sum_{j=1,\ldots P} \beta_j (1-\lambda_i)^{2j} \left|\left|\lambda_i \left(\boldsymbol{U}^{\mathrm{T}}\boldsymbol{H}\right)_i\right|\right|_2^2 \\
&\quad\le \sum_{i=1,\ldots M} \left\|\lambda_i \left(\boldsymbol{U}^{\mathrm{T}}\boldsymbol{H}\right)_i\right\|_2^2.
\end{aligned} \tag{A6}$$

We can obtain

$$\Delta\left(\tilde{\boldsymbol{A}}^P \boldsymbol{H}\right) \le \Delta(\boldsymbol{S}\boldsymbol{H}) \le \Delta(\boldsymbol{H}). \tag{A7}$$

From the above analysis, it can be obtained that the smoothing ability of $\boldsymbol{S}$ is between $\boldsymbol{I}$ and $\tilde{\boldsymbol{A}}^P$. Because β_j is obtained by training, and the smoothing ability of MGCN depends on β_j, the proposed method can adaptively adjust the smoothing ability of the graph convolution, which can avoid undersmoothing and oversmoothing problems to some extent and further improve the classification accuracy.

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